

TECHNOLOGICAL DESIGN

Technological design courses provide students with a variety of learning experiences that focus on the practical application of the principles of engineering, architecture, and design. These activity-based courses emphasize problem solving to meet design challenges in a wide range of areas, which may include apparel and textile design, architectural design, interior design, mechanical and industrial design, and robotics and control systems.

Students learn to apply knowledge of research, historical trends, design, materials, fabrication methods, and testing criteria to develop innovative and environmentally sustainable products, processes, and/or services. The technologies and processes used to create design solutions may include both traditional and computer-based drafting methods, scale models, working prototypes, animations and simulations, displays, portfolios, and presentations.

The list of approved emphasis areas for technological design can be found at www.edu.gov.on.ca/eng/curriculum/secondary/teched.html.

- Courses in technological education are suitable for use in cooperative education programs and in connection with other forms of experiential learning as well as in programs such as the Specialist High Skills Major (SHSM). For more information, see pages 43–44 of this document.
- For policy guidelines pertaining to multiple-credit courses and emphasis courses, see pages 17–18 of this document.

Technological Design in the Twenty-first Century, Grade 12

Open

TDJ40

This course focuses on the relationship between society and technological development. Students will use appropriate tools, techniques, and strategies to research, design, build, and assess prototypes for products and/or processes that respond to society's changing needs. Students will describe how social factors, including culture, media, politics, religion, and environmental concerns, influence technological design. Students will also learn about professional practices in the field, and will research postsecondary pathways leading to careers related to technological design.

Prerequisite: None

A. TECHNOLOGICAL DESIGN FUNDAMENTALS

OVERALL EXPECTATIONS

By the end of this course, students will:

- A1.** describe the design process and ways in which technological design is influenced by societal needs;
- A2.** describe strategies, techniques, and tools for researching, planning, and managing projects to meet a specific societal or other need;
- A3.** demonstrate an understanding of various methods for graphically representing design ideas and solutions;
- A4.** demonstrate an understanding of various kinds of models and prototypes, and describe methods and equipment for making and assessing them;
- A5.** demonstrate an understanding of the terminology and communication methods and formats used in the design process.

SPECIFIC EXPECTATIONS

A1. Design Process

By the end of this course, students will:

- A1.1** describe the purpose of design for a given project (*e.g., technological convergence, cost-efficient products, smaller living spaces*) with reference to key technological concepts (*e.g., aesthetics, control, environmental sustainability/stewardship, ergonomics, fabrication, function, innovation, material, mechanism, power and energy, structure, safety, systems*) (see pp. 7–8);
- A1.2** describe ways in which societal needs, including environmental and economic factors, influence technological design (*e.g., need for products that are smaller, lighter, faster, safer, and/or easier to use*);
- A1.3** describe the steps in the design process (*e.g., define the problem or challenge, taking into account relevant contextual or background information; gather information about criteria, constraints, and available materials; generate possible solutions, using techniques such as brainstorming; choose the best solution; develop and produce a model or prototype; test the model or prototype; incorporate improvements or redesign and retest; report results*) (see pp. 22–23);
- A1.4** describe the relationship between various steps of the design process (*e.g., testing a model or prototype, and then incorporating improvements or redesigning and retesting if necessary*).

A2. Research, Planning, and Organization

By the end of this course, students will:

- A2.1** investigate and describe a variety of strategies, techniques, and tools for gathering pertinent information from appropriate sources (*e.g., magazines covering social issues, Ministry of Culture website, interview with a cultural leader*);
- A2.2** describe strategies for the planning and management of human, material, and financial resources related to a design project (*e.g., delegating tasks, selecting materials, estimating costs*).

A3. Representing Design Ideas Graphically

By the end of this course, students will:

- A3.1** describe different methods for representing design ideas graphically (*e.g., mind maps, sketches, design layouts, computer-aided drafting*) with reference to principles and elements of graphic design (*e.g., principles: contrast, proportion, movement, emphasis, pattern, balance; elements: line, colour, space, shape, size, value*);
- A3.2** demonstrate an understanding of drawing types (*e.g., pictorial and rendered drawings, floor plans, elevations, sections, detail drawings*) and of drafting methods (*e.g., hand drawn, computer-aided*), standards (*e.g., Canadian Standards*).

Association [CSA]), and conventions (e.g., for dimensions, symbols, and abbreviations);

- A3.3** accurately interpret technical drawings and specifications.

A4. Making and Testing Models and Prototypes

By the end of this course, students will:

- A4.1** compare various kinds of models (e.g., conceptual, physical, virtual, theoretical) and prototypes (e.g., proof-of-principle, functional, form study), taking into account the tools, equipment, procedures, and materials required to produce them (e.g., *tools: fabric shears, computer-aided design [CAD] software, computer numerical control [CNC] machine tools; equipment: computer, cut-out pattern; procedures: tracing, forming; materials: matt board, starch, linen, neoprene*);
- A4.2** describe ways to assess models on the basis of various design criteria (e.g., societal needs, functionality, size, weight, durability, aesthetics, use of recyclable materials, cultural appropriateness);
- A4.3** identify and describe equipment (e.g., decibel meter, height gauge, compression tester) for assessing prototypes.

A5. Reporting and Presenting

By the end of this course, students will:

- A5.1** demonstrate an understanding of industry-standard terminology (e.g., *scale versus ruler, artistic drawing versus technical drawing*);
- A5.2** investigate and compare a variety of presentation methods (e.g., *oral report, multimedia presentation, use of presentation software, use of overhead versus LCD projector, multimedia versus slides*) based on relevant criteria (e.g., *cost, ease of use, effectiveness for delivering intended message*);
- A5.3** describe the characteristics of various styles and formats for tracking and reporting the progress of a project (e.g., *journals, portfolios, blogs*).

B. TECHNOLOGICAL DESIGN SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** use appropriate strategies and tools to research, plan, organize, and manage design projects;
- B2.** develop designs and represent them graphically, using industry standards for illustrations and technical drawings;
- B3.** construct models and prototypes, using a variety of techniques, tools, and materials, and assess these models and prototypes in terms of the design criteria;
- B4.** report on design choices, the societal influences that helped determine those choices, project evaluations, and reflections on their experiences throughout the design process, using appropriate formats and styles.

SPECIFIC EXPECTATIONS

B1. Research, Planning, and Organization

By the end of this course, students will:

- B1.1** gather and analyse relevant background information and design ideas from various sources (*e.g., Canadian Standards Association [CSA] publications, Ontario Building Code, Machinery's Handbook*);
- B1.2** investigate and report on societal factors that influence technological design (*e.g., news media, politics, religion, environment, gender, cultural and ethnic diversity*);
- B1.3** use appropriate strategies and tools (*e.g., sequence chart, time sheet, checklist, file management*) to help plan, organize, and manage design projects;
- B1.4** research and identify relevant design criteria and constraints relating to societal influences (*e.g., cultural considerations when designing restaurants, use of environmentally friendly materials, gender-specific designs*).

B2. Developing and Representing Design Ideas

By the end of this course, students will:

- B2.1** create and present freehand sketches to illustrate initial design concepts, applying a variety of elements and principles of graphic design;

- B2.2** apply mathematical and scientific concepts and skills as required in the course of designing various products and/or processes;

- B2.3** produce technical drawings (*e.g., orthographic projection, isometric projection*) and/or presentation-quality drawings (*e.g., perspective renderings, 3D computer imaging*) of design solutions, using common drafting standards (*e.g., line types, dimensioning, title block*).

B3. Making and Testing Models and Prototypes

By the end of this course, students will:

- B3.1** use appropriate tools, equipment, and materials to create design models and prototypes (*e.g., tools: rotary tool, bandsaw, fabric shears, computer-aided design [CAD] software; equipment: computer, sewing machine; materials: paper, wax, clay, wood, metals, composites, plastics, cardboard, starch, textiles*);
- B3.2** use appropriate metric and imperial measuring tools, scales (*e.g., metric: 1:10, 1 cm:1 m or 1:100, 1:500; imperial: 1/2":1' or 1:24*), and proportion techniques when creating a model or prototype;
- B3.3** analyse products and/or processes in terms of established criteria and constraints (*e.g., appearance, durability, ease of use, cost, budgetary constraints, societal concerns and preferences*).

B4. Reporting and Presenting

By the end of this course, students will:

B4.1 create and present reports summarizing design choices and results of evaluations of projects, using a variety of appropriate formats (*e.g., multimedia or oral presentation, technical report*), with an emphasis on how the design was affected by societal influences;

B4.2 report and reflect on their experiences throughout the design process, assessing individual skills development and the productivity and dynamics of the team, using appropriate written and/or oral formats.

C. TECHNOLOGY, THE ENVIRONMENT, AND SOCIETY

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** demonstrate an understanding of environmentally responsible design practices and strategies, and apply them in the technological design process and related activities;
- C2.** research and describe the relationship between society and technological development.

SPECIFIC EXPECTATIONS

C1. Technology and the Environment

By the end of this course, students will:

- C1.1** demonstrate an understanding of environmental issues that affect product design (*e.g., finite non-renewable resources, pollution, wastes, greenhouse gases, climate change, ozone depletion, life cycle of products*);
- C1.2** describe, promote, and apply design practices that conserve energy and other resources (*e.g., reuse or recycle lumber and other materials, use materials with recycled content, use wood glue instead of hot glue, use energy-management software for computers and other electronic equipment, use renewable energy, use high-efficiency motors and appliances*);
- C1.3** demonstrate an understanding of technological, political, and social strategies for managing the relationship between society and the environment (*e.g., technological developments to improve energy efficiency and/or reduce emissions, lobbying governments for regulations and/or funding to improve the environment, education about environmental issues*);
- C1.4** explain how good design can reduce the wastes produced by the manufacture and use of products.

C2. Technology and Society

By the end of this course, students will:

- C2.1** research and report on the history of technological eras (*e.g., agricultural, industrial, and information eras*) and developments (*e.g., invention of seed drill, mechanization of production, wireless communication*);
- C2.2** describe how society influences technology (*e.g., higher energy costs spur development of more efficient vehicles, increasing population density leads to the construction of taller buildings, environmental awareness leads to development of alternative energy sources*);
- C2.3** describe how technological change affects society (*e.g., health care developments increase lifespan, robotics replace human resources*);
- C2.4** describe intellectual property rights related to the design of products (*e.g., copyright, trademark, patent*).

D. PROFESSIONAL PRACTICE AND CAREER OPPORTUNITIES

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** describe and apply appropriate health, safety, and environmental practices and standards throughout the design process;
- D2.** investigate and describe a variety of careers related to technological design, and describe the training and education required for them.

SPECIFIC EXPECTATIONS

D1. Health and Safety

By the end of this course, students will:

- D1.1** describe health and safety laws, regulations, standards, and agencies that relate to technological design (*e.g., Ontario Building Code, Canadian Standards Association [CSA], Ministry of Labour*);
- D1.2** adhere to personal and environmental health and safety standards and procedures with respect to processes, materials, tools, equipment, and facilities throughout the design process and related activities (*e.g., use protective equipment; set tool and equipment guards properly; ensure adequate ventilation and ergonomic seating and other workplace arrangements; follow safe operating procedures; keep work areas clean and organized; store materials and dispose of wastes properly*);
- D1.3** use protective clothing, gear, and equipment appropriately (*e.g., dust mask, safety glasses*);
- D1.4** describe the rights and responsibilities of employees under the Occupational Health and Safety Act (*e.g., right to know, right to refuse, right to participate*).

D2. Career Opportunities

By the end of this course, students will:

- D2.1** describe a variety of career and business opportunities related to technological design (*e.g., engineering, architecture, industrial design, control systems, apparel/textile design*);
- D2.2** identify and compare the educational and training requirements (*e.g., degree, diploma, certificate, apprenticeship*) for specific careers in technological design and related trades and services;
- D2.3** identify groups and programs that are available to support students who are interested in pursuing non-traditional career choices in the technological design industry (*e.g., mentoring programs, virtual networking/support groups, specialized postsecondary programs, relevant trade/industry associations*);
- D2.4** demonstrate an understanding of and apply the Essential Skills that are important for success in the technological design industry, as identified in the Ontario Skills Passport (*e.g., reading text, writing, document use, computer use, oral communication, numeracy, thinking skills*);
- D2.5** demonstrate an understanding of and apply the work habits that are important for success in the technological design industry, as identified in the Ontario Skills Passport (*e.g., working safely, teamwork, reliability, organization, working independently, initiative, self-advocacy*);
- D2.5** maintain an up-to-date portfolio that includes pieces of work and other materials that provide evidence of their skills and achievements in technological design (*e.g., work logs, skills checklist, sketches, drawings, photographs of models and prototypes, virtual models*), and explain why having a current portfolio is important for career development and advancement.